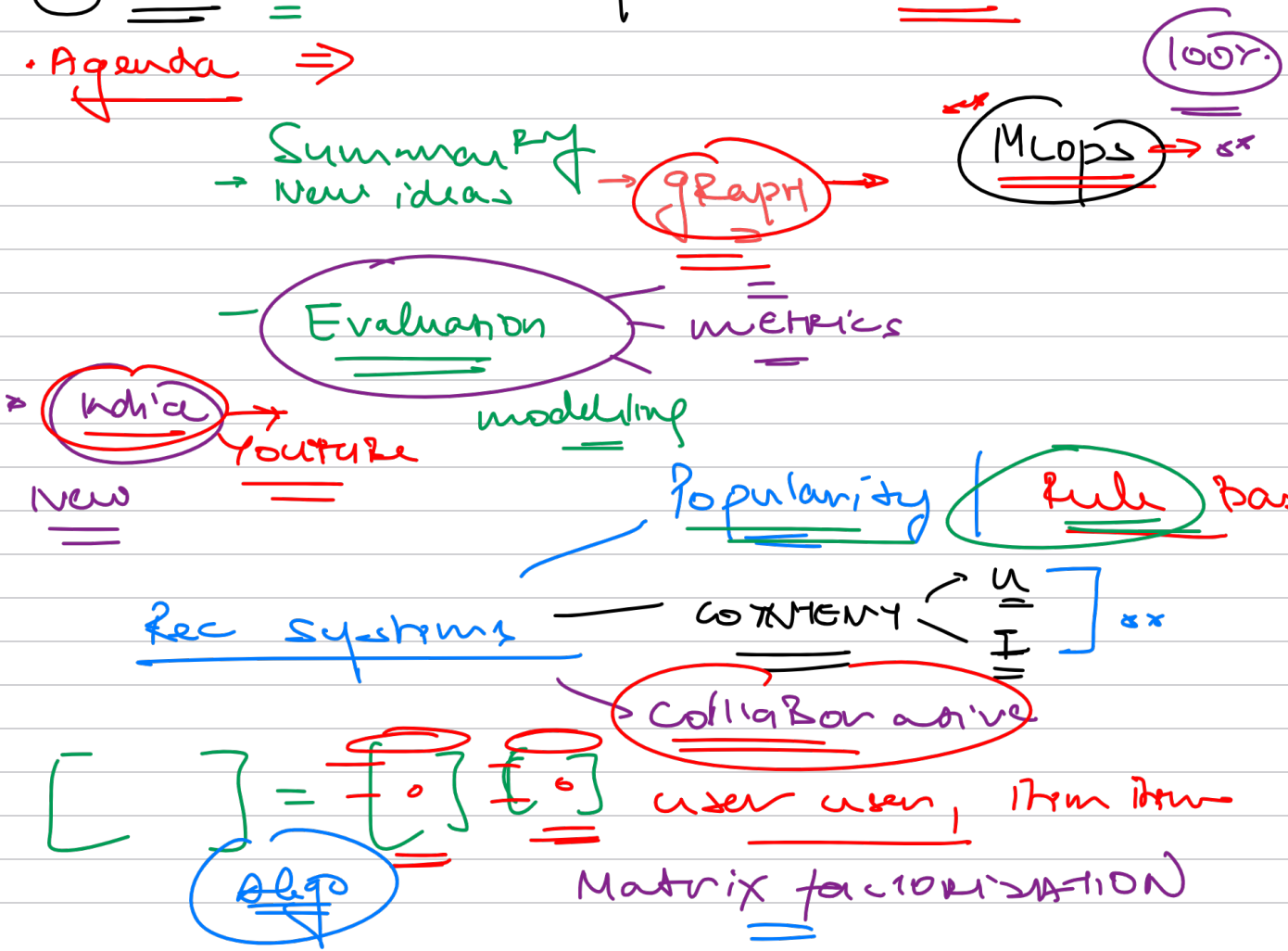
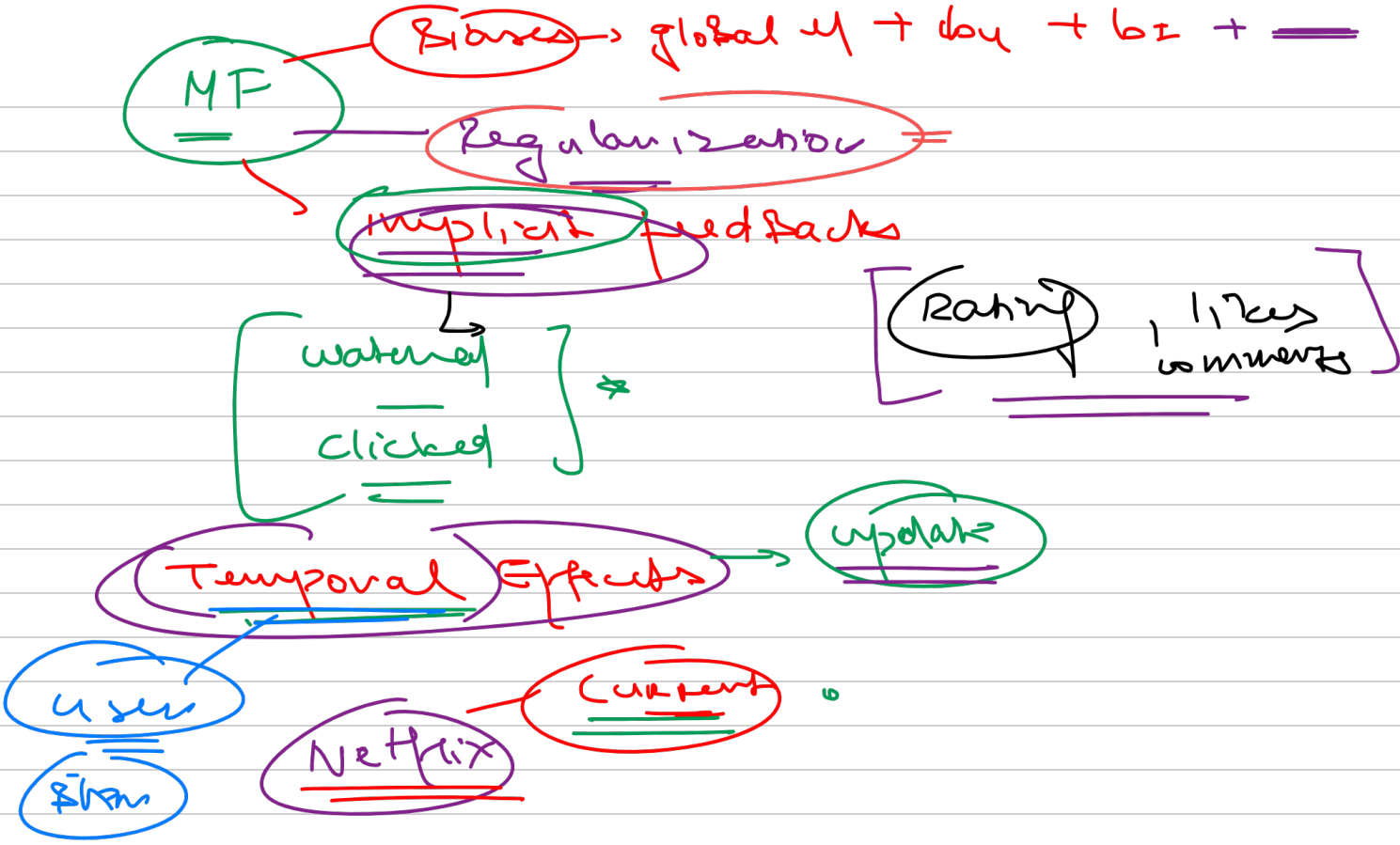
Recommender Systems END ⇒

- Agenda ⇒




Biases → global μ + bu + bi +
MF
Regularization
Implicit feedbacks
watched
clicked
Rating
likes
comments
Temporal Effects
popularity
User
Bias
Netflix
Current

User — Items

Netflix

User → movies

Amazon

Spotify

facebook / LinkedIn

* (People u may know)

User — user

Items

② Cold START

Interaction history

Collab ✗

① New user

Netflix → ratings

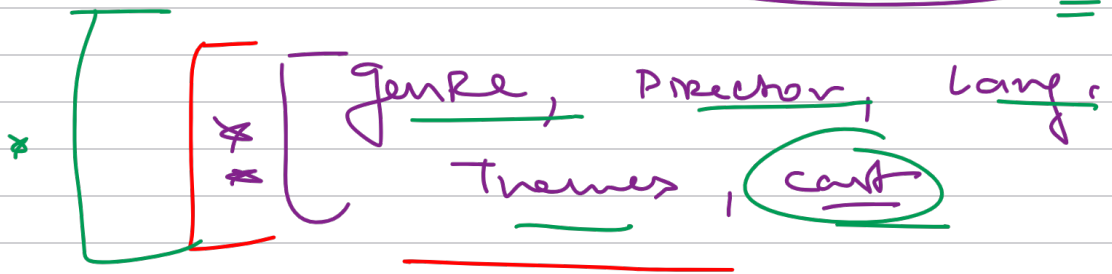
watch history

② New item

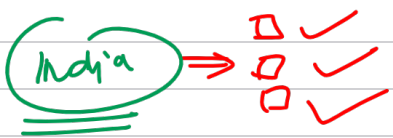
collab ⇒

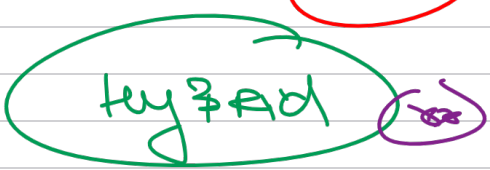
→ CONTENT Based

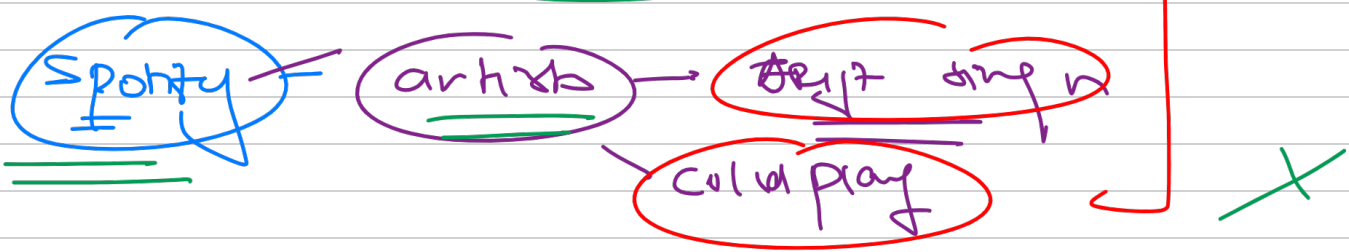
New movie = Dune Part 3



Users → similar content







(Users initial interests)

(Popularity | Generalized Rec)

Trending in India

Web — Youtube — most watched

popular in tech

useful fallback

Rule

Youtube — (Top 10 Trending in India)

× Views + Recent growth

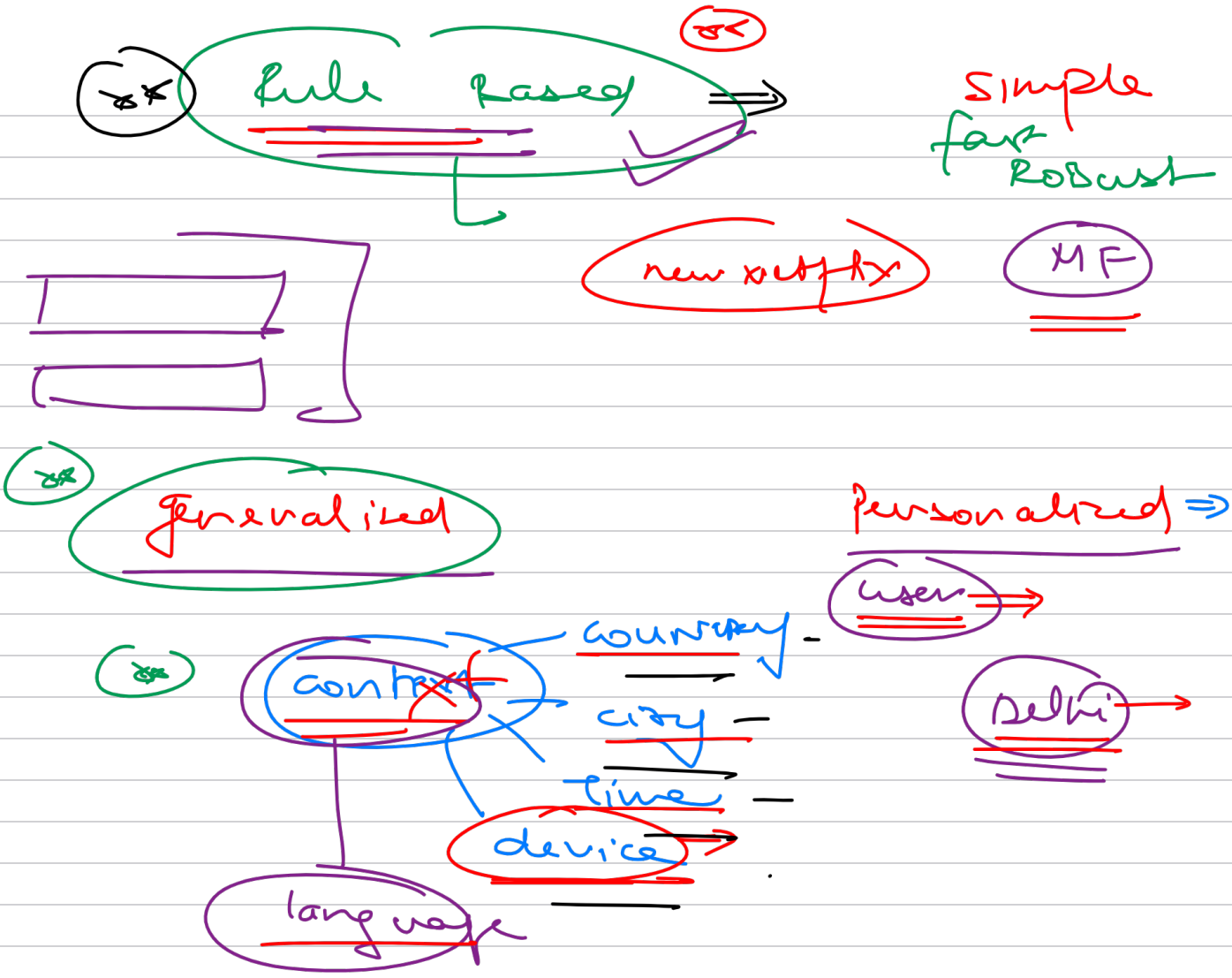
Geography

+ Recency

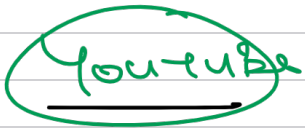
Most viewed

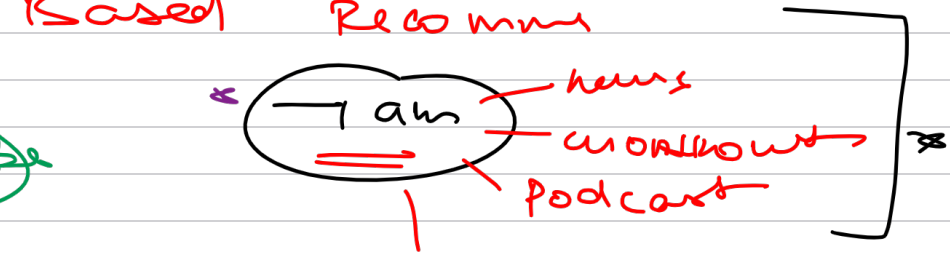
Breaking news

Trending songs

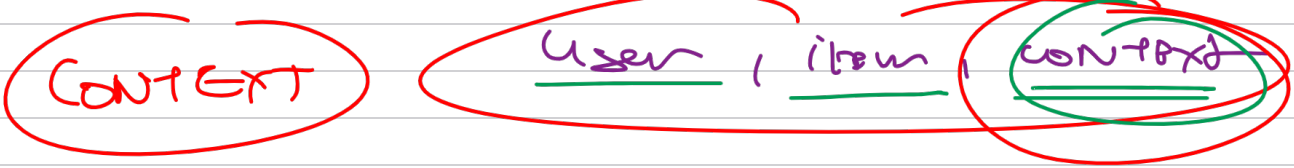

Rule Based
simple
for
Robust
new
MF
Generalized
Personalized
User
context
country
city
time
device
Delhi
language

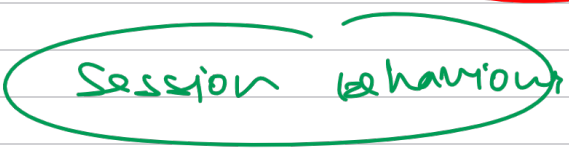
CONTEXT Based Recommn

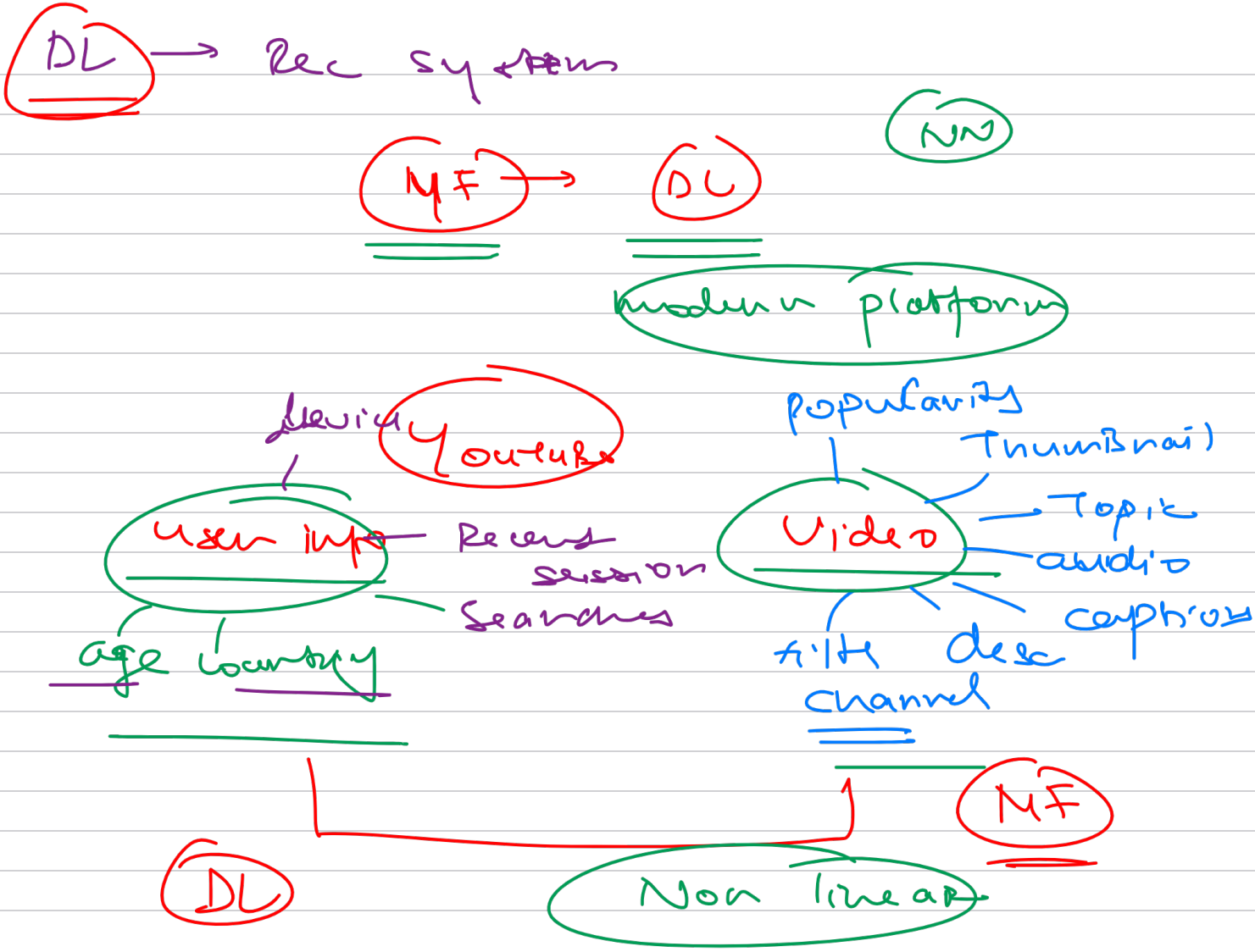


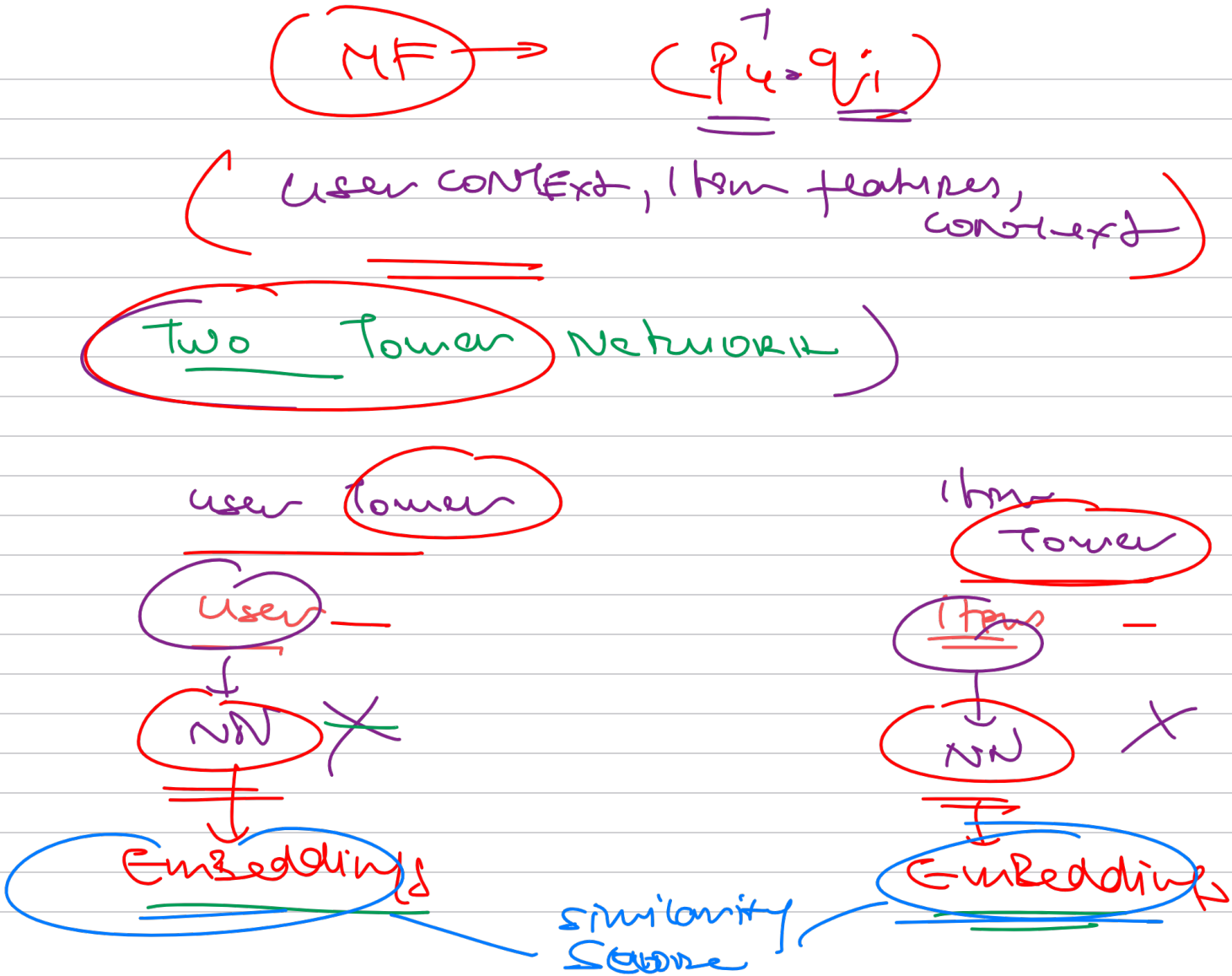


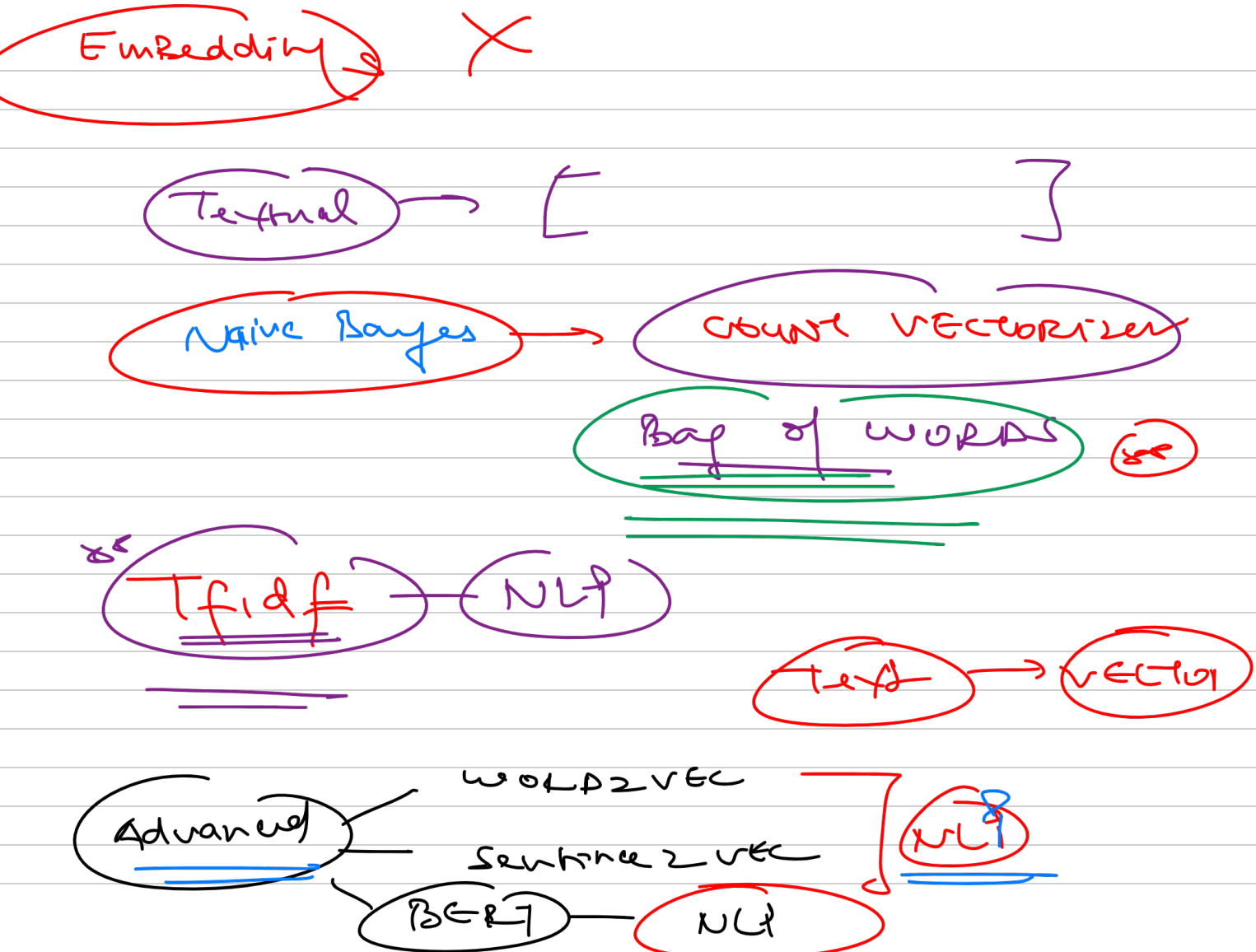
User item






DL
Rec system
NN
MF
DL
modern platform
Youtube
device
user info
Recent
session
Searches
age
country
popularity
Thumbnail
Video
Topic
audio
caption
title
desc
channel
MF
DL
Non linear


MF
$p_u \cdot q_i$
User context, Item features, context
Two Tower Network
User Tower
User
NN
Embeddings
Item Tower
Item
NN
Embeddings
similarity score


Embedding
Textual
Naive Bayes
Count Vectorizer
Bag of words
Tfidf
NLP
text
vector
Advanced
Word2Vec
Sentence2Vec
BERT
NLP
NLP

facebook

"You may know"

Alex

James

Adams

Robert

Solution ⇒

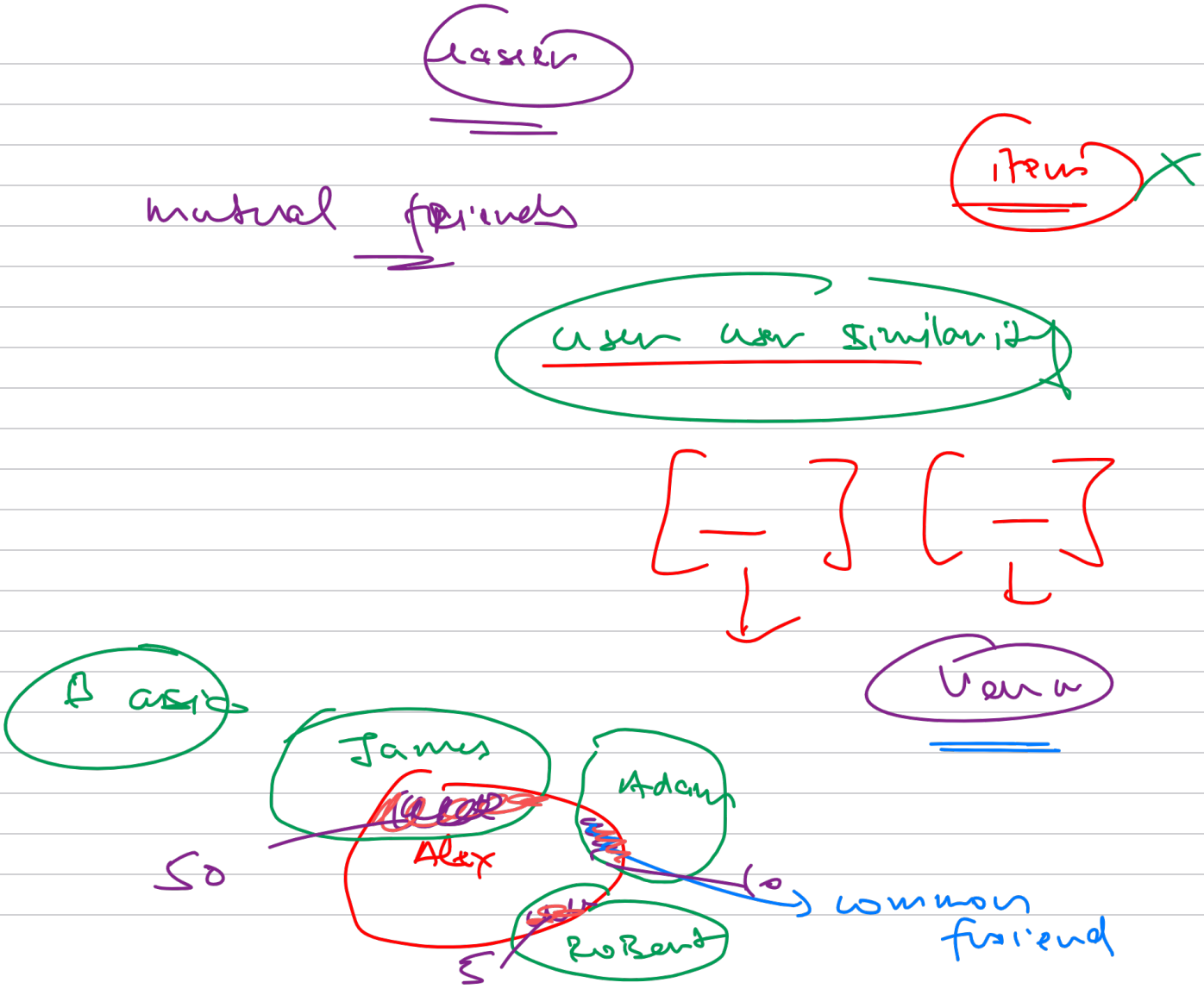
data ?

→ Demographics

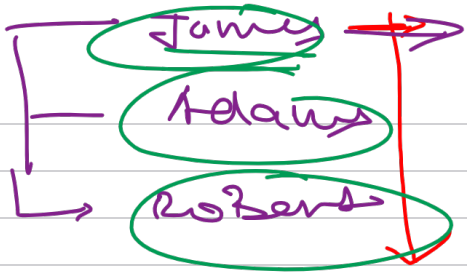
age, sex, loc^n

education, college

+ workplace

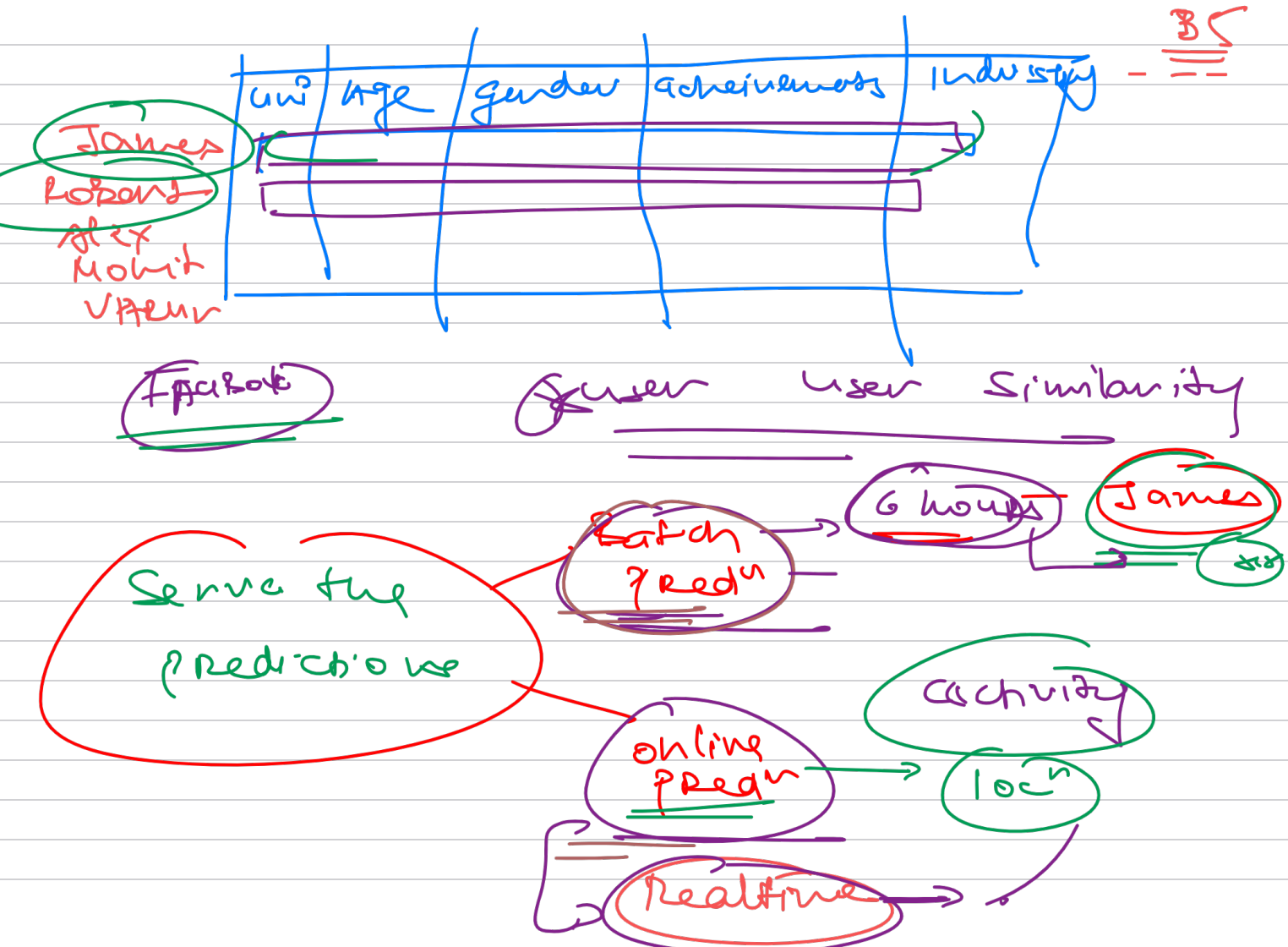

Users
Items
mutual friends
user user Similarity
Users
James
Adam
Alex
Robert
So
common friend

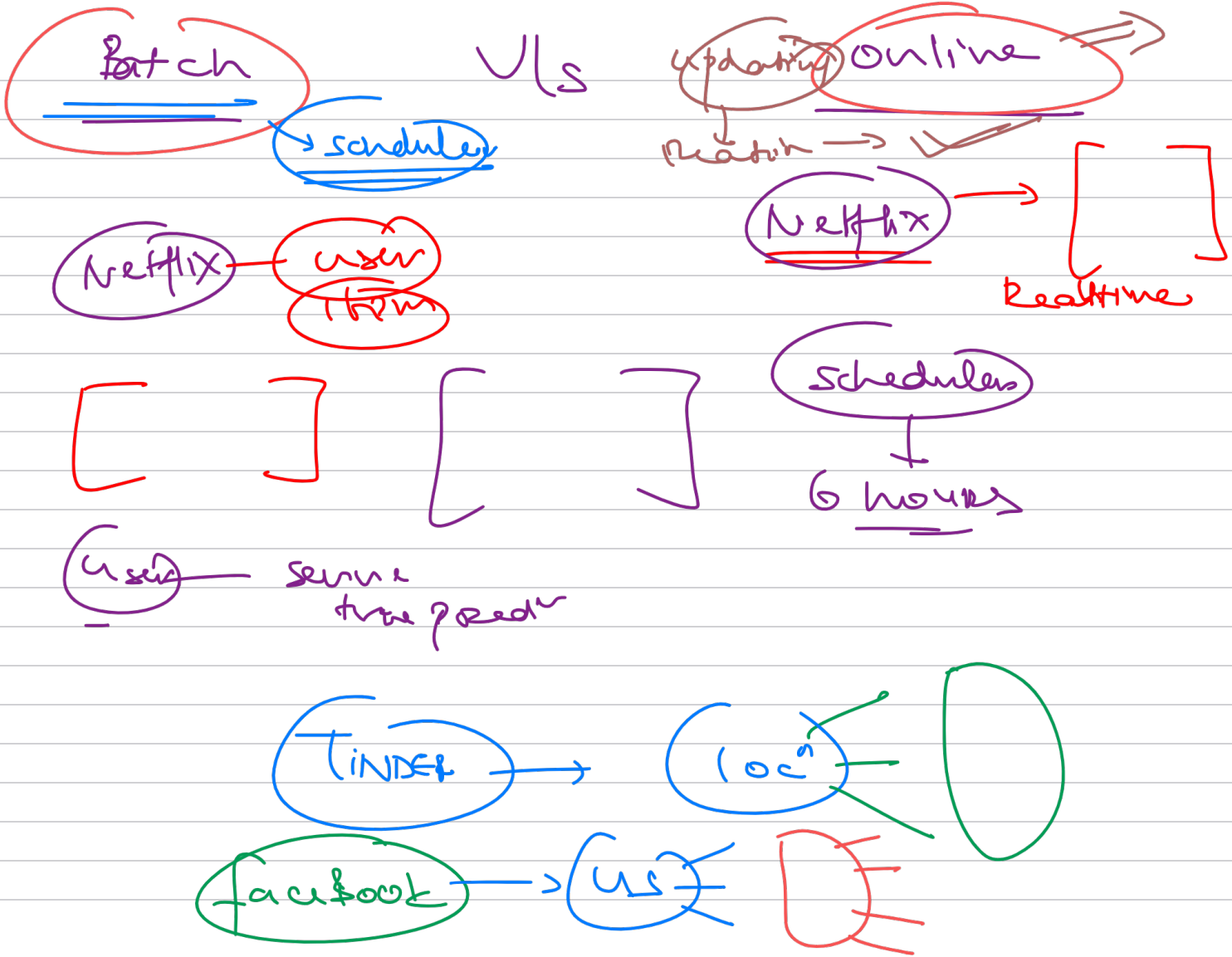
You may know →




```
for u in users
    if len(u.friends.intersection(my friends)) > Threshold
        Recommend = True
        Recommend.append(u)
```

⇒ Mutual friends

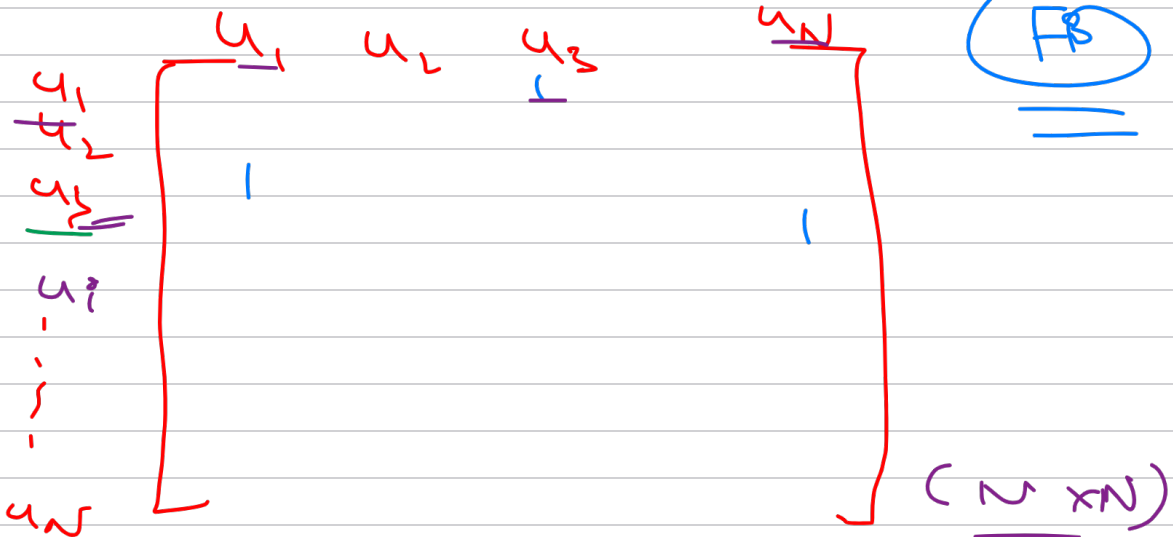

BC
uid
age
gender
achievements
industry
James
Robert
Alex
Mohit
Vikram
Facebook
User
User Similarity
Serve the Predictions
Batch Predn
6 hours
James
Activity
Loc^n
online Predn
Realtime


Batch
VIs
scheduler
updating
online
Netflix
user
Netflix
Real time
Scheduler
6 hours
User
Server
facebook
US

Matrix

User users

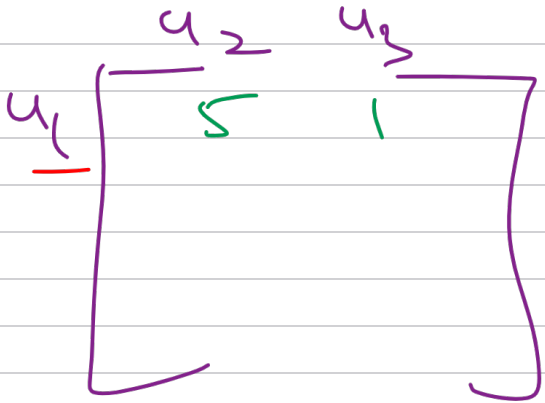
FB

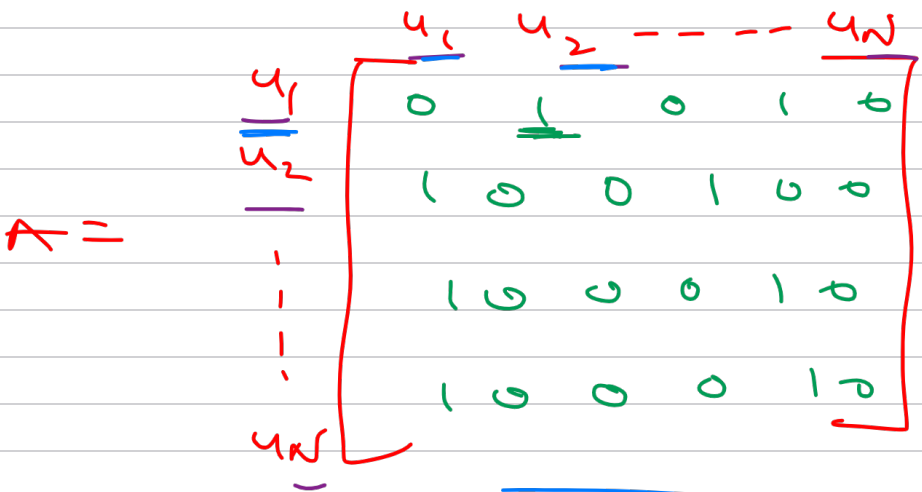


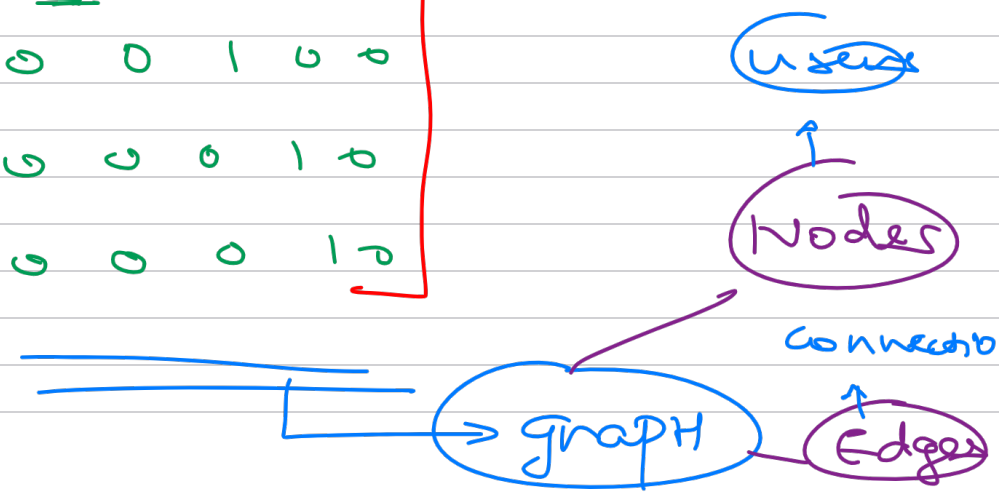
(1) how much they have interacted with each other

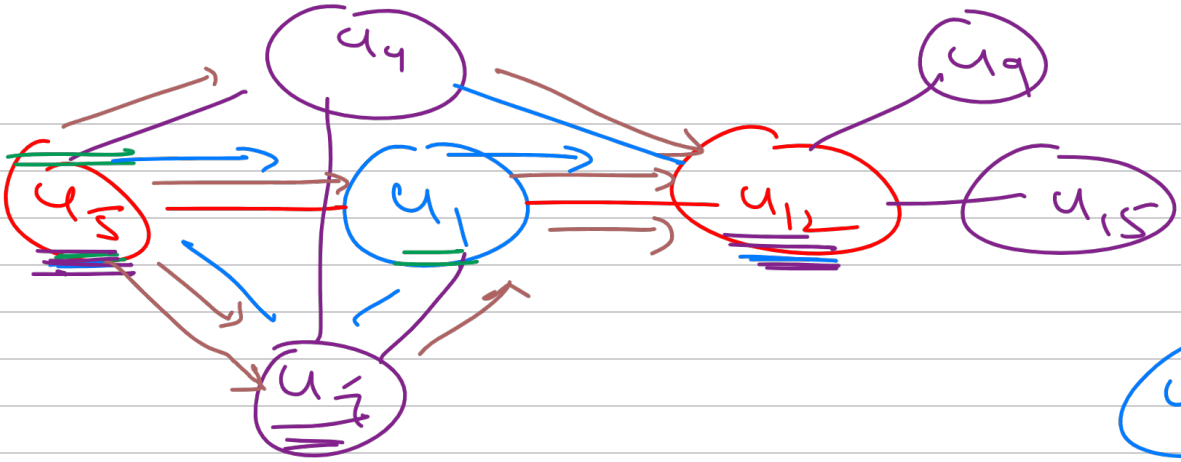
liked posts, comments, tagged

Strength ⟹ 1 − S


u_2
u_3
u_1
5
1

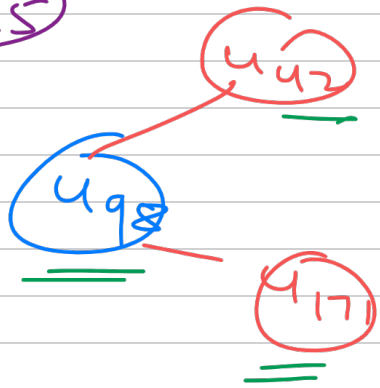

u_1
u_2
u_N
u_1
u_2
u_N
A =
0 1 0 1 0
1 0 0 1 0 0
1 0 0 0 1 0
1 0 0 0 1 0


Users
Nodes
connection
graPH
Edges



geography

workspace









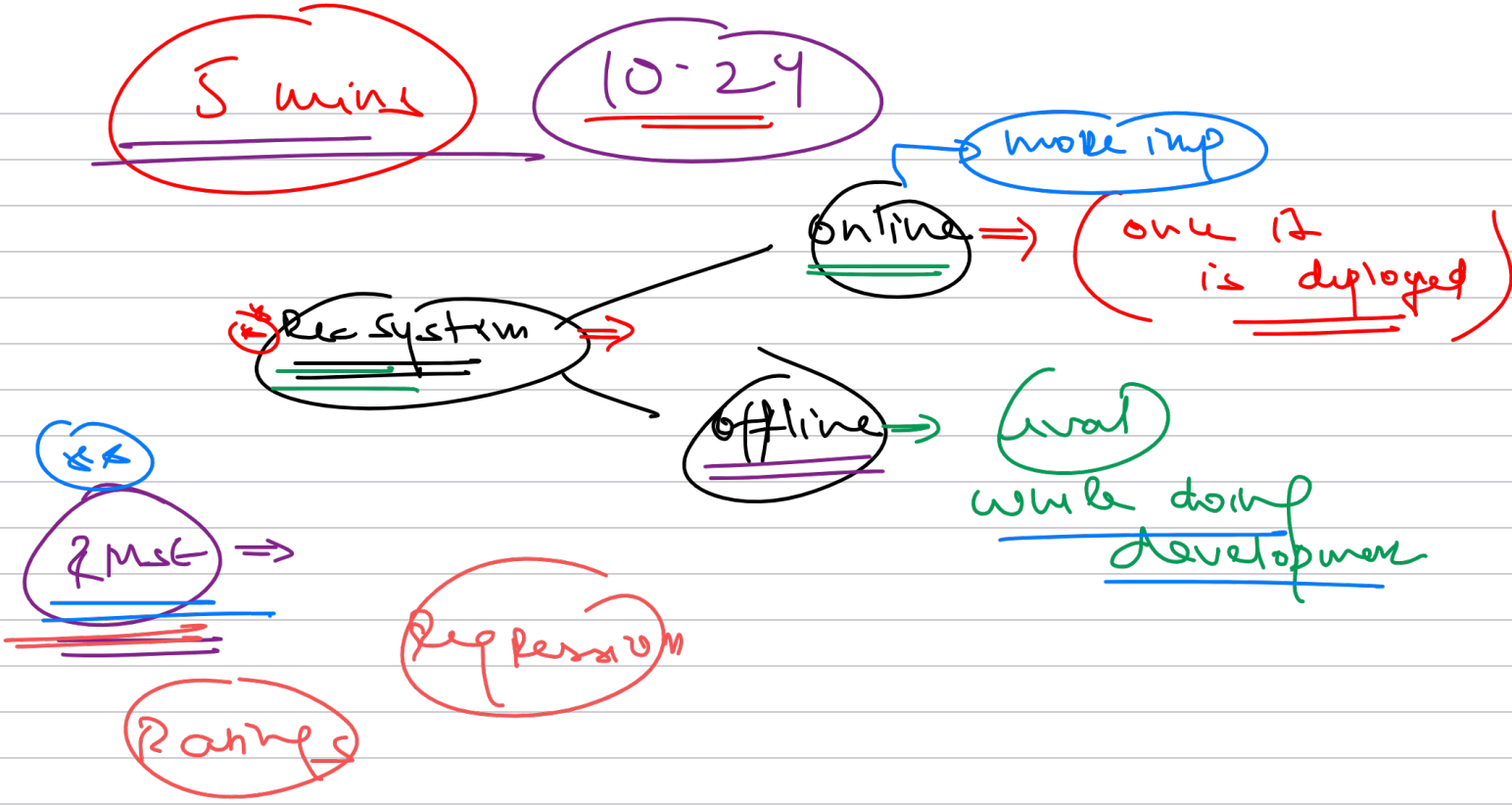
connection



multiple paths

= more likelihood

networkx

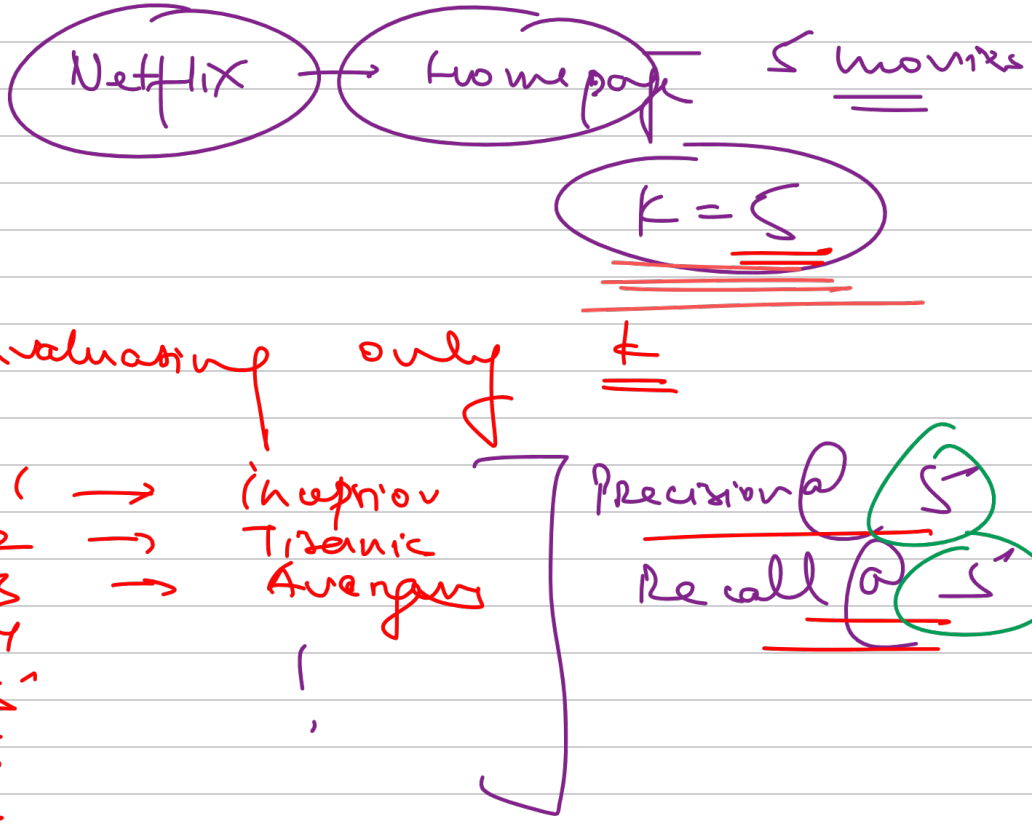


Precision@K: "Out of the K items I recommended, how many were actually relevant?"

Recall@K: "Out of all the relevant items available for this user, how many did I manage to recommend?"

NDCG@K: "Did I put the most relevant items near the top of the recommendation list?"

@ K


Netflix
Homepage
5 movies
K = 5
evaluating only for
1 → Inception
2 → Titanic
3 → Avengers
4
5
6
Precision @ 5
Recall @ 5

Precision @ K

of Items Recommended

How many are Relevant

Precision @ K = $\frac{\text{Relevant items in top K}}{K}$

Amazon

Recommender system

User

1 — Laptop
2 — Mouse
3 — Keyboard
4 — Monitor
5 — Headphones

Relevant

✓ -
✗ -
✓ -

Precision@ $k = \frac{3}{5} = 0.6$

$= 60\%$

Precision

Spotify — 10 songs

2 songs

Precision $= \frac{2}{10} = 20\%$

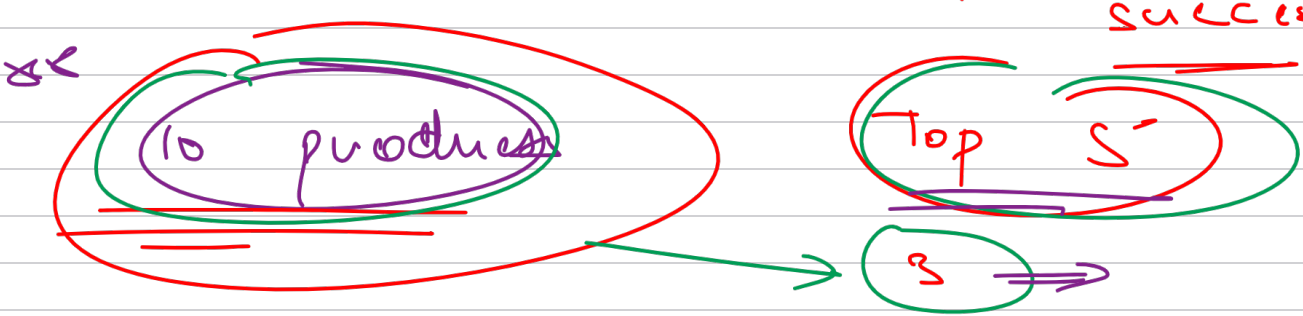
Recommendation quality

Relevancy — Click, Bought

Recall @ k

Out of all Relevant items for this user,

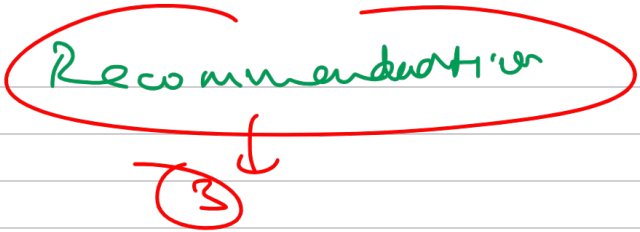
how many did we Retrieve successfully



$$\text{Recall @ 5} = \frac{3}{10} = 0.3 = 30\%.$$

$$\text{Recall @ k} = \frac{\text{Relevant items in Top k}}{\text{Total Relevant items for user}}$$

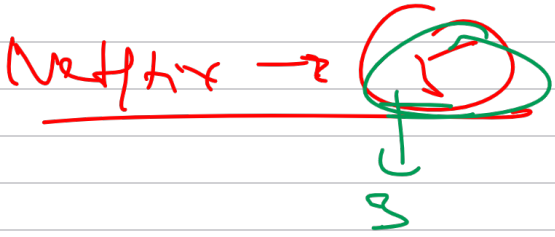




$$\text{Recall} = \frac{3}{10} = 30\%$$







$$\text{Precision} = \frac{3}{5} = 60\%$$

$$\text{Recall} = \frac{3}{10} = 30\%$$

Total Relevant items

A B C D E F G H I J

10

Top 5 Rec

A
B
X
C
Y

Relevant = 3

Precision = 3/5

Recall = 3/10

Google search ⇒ Best ML interview ques

Google → 10 → 8 → useful

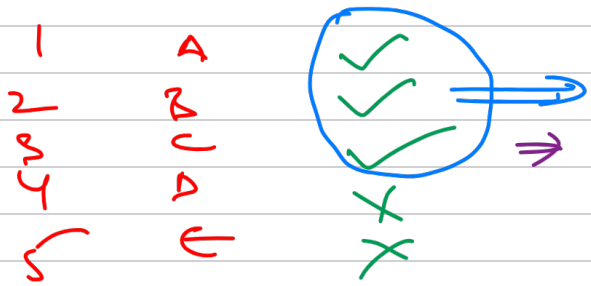
Precision = 8/10

100 →

Recall = 8/100

Top 2 metrics

System A



System B

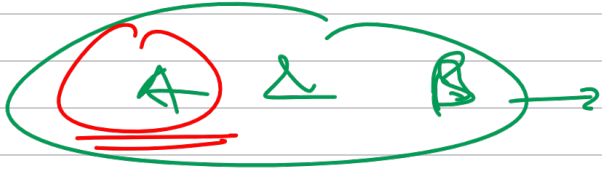


Ranking

to → Relevant

Precision = 3/5

Recall = 3/10



→ identical

NDCG

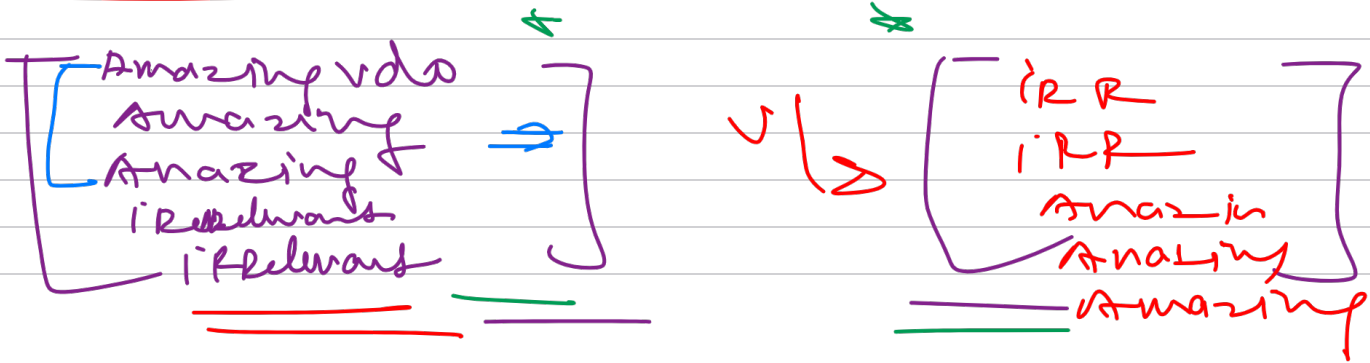
Normalized Discounted Cumulative gain

Reward → Relevant Recs

Relevant Item → Rank 1

Rank 20

YouTube Rec



DCG

Gain

Relevant
rel → Reward

DISCOUNT

Reduce Reward

Binary

Relevant → 1
NR → 0

$$DCG@k = \sum_{i=1}^{k} \frac{rel_i}{\log_2(i+1)}$$

Rank 1 → almost full Reward

Rank 2 → smaller Reward

Rank	Item	Relevant
1	A	1
2	B	0
3	C	1
4	D	0
5	E	1

$\frac{1}{\log_2(2)} = 1$

$\frac{1}{\log_2(4)} = 0.5$

$\frac{1}{\log_2(6)} = 0.387$

DCG@5

Rank 1

DCG → 1 + 0.5 + 0.387

= 1.887

Best possible ranking

IDCG = ideal DCG

Rank	Item	Relevant
1	A	1
2	B	0
3	C	1
4	D	0
5	E	1

ideal DCG

$1 + \frac{1}{\log_2(3)} + \frac{1}{\log_2(4)}$

$= 1 + 0.631 + 0.5$

$= 2.131$

ideal DCG

NDCG

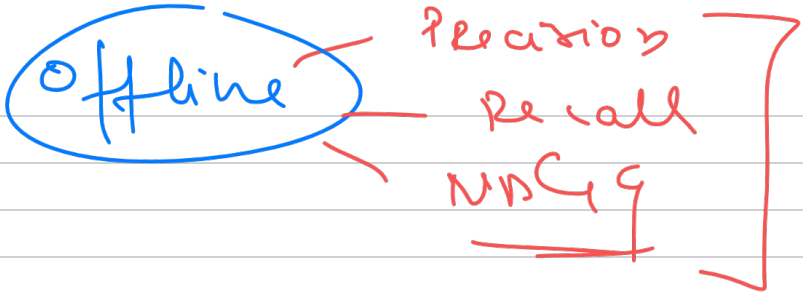
NDCG

$$NDCG = \frac{DCG@k}{idcg@k}$$

$$= \frac{1887}{2131} = 0.885$$

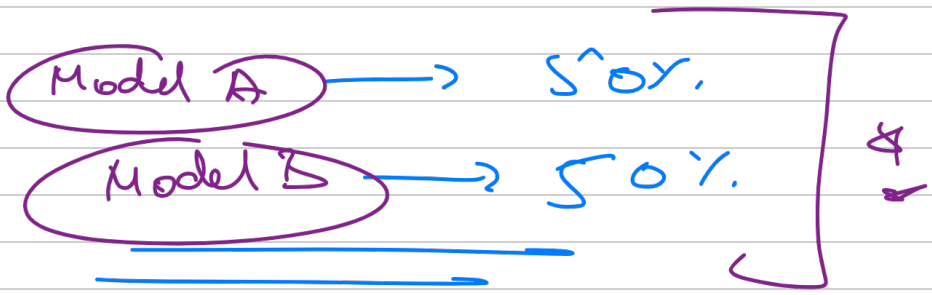
NDCG → [0, 1]

0 ↙


Offline
Precision
Recall
NDCG


Online
Clicks
DAU

A | B Testing


Model A
50%.
Model B
50%.

Netflix / YT

- CTR
- watch time
- completion Rate
- Retention

Amazon

- Add To Cart Rate
- conversion
- Revenue

offline VIs online

Spotify

- Song completion
- skip rate
- Saves
- listening time